where the right-hand side is

$$\begin{aligned}
&= \tfrac{1}{2}\mathrm{a}_1 \cdot \xi_3 + \eta_3 \cdot \zeta\zeta \\
&= (\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2)(-(\tfrac{1}{2}\xi_2 + \eta_2 \cdot \zeta\zeta)) \\
&\quad + (\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3)(\xi_1^2 + \eta_1^2 + \zeta_1^2),
\end{aligned}$$

which is

$$= G \cdot \tfrac{1}{2}\mathrm{a}_1 + F(\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2) + C(\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3),$$

or, as this might be written,

$$= (G,\ F,\ C \,\langle\, \tfrac{1}{2}\mathrm{a}_1,\ \mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2,\ \mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3 \rangle).$$

Retaining the former form, the result is

$$L\square = G \cdot \tfrac{1}{2}\mathrm{a}_1 + F(\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2) + C(\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3),$$

and it is now very easy to write down the complete system of the 18 equations; viz. if the determinant above written down be called 11.1.2, and so in other cases, then we have

	2.3	3.1	1.2
	A	H	G
	H	B	F
	G	F	C
$L.11$	$\tfrac{1}{2}\mathrm{a}_1$	$\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2$	$\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3$
$.\,22$	$\mathrm{h}_2 - \tfrac{1}{2}\mathrm{b}_1$	$\tfrac{1}{2}\mathrm{b}_2$	$\mathrm{f}_2 - \tfrac{1}{2}\mathrm{b}_3$
$.\,33$	$\mathrm{g}_3 - \tfrac{1}{2}\mathrm{c}_1$	$\mathrm{f}_3 - \tfrac{1}{2}\mathrm{c}_2$	$\tfrac{1}{2}\mathrm{c}_3$
$.\,23$	$-\tfrac{1}{2}(-\mathrm{f}_1 + \mathrm{g}_2 + \mathrm{h}_3)$	$\tfrac{1}{2}\mathrm{c}_1$	$\tfrac{1}{2}\mathrm{b}_3$
$.\,31$	$\tfrac{1}{2}\mathrm{c}_2$	$\tfrac{1}{2}(\mathrm{f}_1 - \mathrm{g}_2 + \mathrm{h}_3)$	$\tfrac{1}{2}\mathrm{a}_3$
$.\,12$	$\tfrac{1}{2}\mathrm{b}_3$	$\tfrac{1}{2}\mathrm{a}_2$	$\tfrac{1}{2}(\mathrm{f}_1 + \mathrm{g}_2 - \mathrm{h}_3)$

read for instance

$$L.11.1.2 = G \cdot \tfrac{1}{2}\mathrm{a}_1 + F(\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2) + C(\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3),$$

the equation obtained above.

There are eighteen functions as shown by the following diagram:

	2.3	3.1	1.2
$(22)(33) - (23)^2$		2.3	
$(33)(11) - (31)^2$		3.1	
$(11)(22) - (12)^2$		1.2	
$(31)(12) - (11)(23)$		2.3	
$(12)(23) - (22)(31)$		3.1	
$(23)(31) - (33)(12)$		1.2,	

viz. in any line of the diagram the bracketed duads may belong to any one at pleasure, but all to the same, of the three pairs 2.3, 3.1, 1.2; thus the first line might be

$$(22.3.1)(33.3.1) - (23.3.1)^2,$$

or, instead of the 3.1, we might have 2.3 or 1.2. But of the 18 functions I distinguish 6, viz. those in which for the six lines respectively the pairs are 2.3, 3.1, 1.2, 2.3, 3.1, 1.2, as shown in the diagram. Each of these six functions can be obtained under two different forms, and by equating these we have the equations $(\mathfrak{A})$, $(\mathfrak{B})$, $(\mathfrak{C})$, $(\mathfrak{f})$, $(\mathfrak{g})$, $(\mathfrak{h})$, before referred to; thus $(\mathfrak{C})$ is obtained by equating two different forms of the function $(11.1.2)(22.1.2) - (12.1.2)^2$, and $(\mathfrak{f})$ is obtained by equating two different forms of the function $(23.1.2)(31.1.2) - (33.1.2)(12.1.2)$.

The determinants 11.1.2, &c., may be denoted by accented letters a, b, c, f, g, h, as follows:

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}, \; \ldots \; = \mathrm{a}', \; \mathrm{b}', \; \mathrm{c}', \; \mathrm{f}', \; \mathrm{g}', \; \mathrm{h}',$$

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_3 & y_3 & z_3 \\ x_1 & y_1 & z_1 \end{vmatrix}, \; \ldots \; = \mathrm{a}'', \; \mathrm{b}'', \; \mathrm{c}'', \; \mathrm{f}'', \; \mathrm{g}'', \; \mathrm{h}'',$$

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}, \; \ldots \; = \mathrm{a}''', \; \mathrm{b}''', \; \mathrm{c}''', \; \mathrm{f}''', \; \mathrm{g}''', \; \mathrm{h}''';$$

viz.

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \mathrm{a}', \quad \&\text{c}.$$

In this notation, $(\mathfrak{C})$ is obtained by equating two values of $\mathrm{a}''\mathrm{b}'' - (\mathrm{h}'')^2$, and $(\mathfrak{f})$ by equating two values of $\mathrm{f}'\mathrm{f}' - \mathrm{g}''\mathrm{h}''$.

The forms which I call the second are those given by the immediate substitution of the foregoing values of (11.1.2), &c.; to obtain the first forms, I proceed to calculate those of $(\mathfrak{C})$ and $(\mathfrak{f})$.

Forming by the ordinary rule the product of the determinants (11.1.2) and (22.1.2), which are

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \quad \text{and} \quad \begin{vmatrix} x_{22} & y_{22} & z_{22} \\ x_1 & y_1 & z_1 \\ x_3 & y_3 & z_3 \end{vmatrix},$$

this is

$$\begin{vmatrix} 11.22 & 1.11 & 2.11 \\ 1.22 & 1.1 & 1.2 \\ 2.22 & 1.2 & 2.2 \end{vmatrix},$$

where 11.22 denotes $x_{11}x_{22}+y_{11}y_{22}+z_{11}z_{22}$, and the like for the other symbols. In like manner, the square of the determinant (12.1.2), that is, of

$$\begin{vmatrix} x_{12} & y_{12} & z_{12} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix},$$

is

$$\begin{vmatrix} 12.12 & 1.12 & 2.12 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 2.1 & 2.2 \end{vmatrix},$$

or, observing that in the two resulting determinants the terms 11.22 and 12.12 are multiplied by the same factor, the expression for the difference gives

$$\begin{aligned} &(11.1.2)(22.1.2) - (12.1.2)^2 \\ &\quad = (11.22 - 12.12)\begin{vmatrix} 1.1 & 1.2 \\ 1.2 & 2.2 \end{vmatrix} + \begin{vmatrix} & 1.11 & 2.11 \\ 1.22 & 1.1 & 1.2 \\ 2.22 & 1.2 & 2.2 \end{vmatrix} - \begin{vmatrix} & 1.12 & 2.12 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 1.2 & 2.2 \end{vmatrix}, \end{aligned}$$

containing $11.22 - 12.12$, which, by what precedes, is

$$= -\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12});$$

the other terms are also known, viz. the whole value is

$$\begin{aligned} &= -\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12})(\mathrm{ab} - \mathrm{h}^2) \\ &\quad + \mathrm{a}\begin{vmatrix} 0 & \frac{1}{2}\mathrm{b}_1 & \frac{1}{2}(\mathrm{b}_2 - \mathrm{h}_1) \\ \frac{1}{2}\mathrm{a}_2 - \mathrm{h}_1 & \mathrm{a} & \mathrm{h} \\ \frac{1}{2}\mathrm{b}_2 - \mathrm{h}_1 & \mathrm{h} & \mathrm{b} \end{vmatrix} - \mathrm{h}\begin{vmatrix} 0 & \frac{1}{2}\mathrm{a}_1 & \frac{1}{2}\mathrm{a}_2 \\ \frac{1}{2}\mathrm{b}_1 & \mathrm{a} & \mathrm{h} \\ \frac{1}{2}\mathrm{b}_2 - \mathrm{h}_1 & \mathrm{h} & \mathrm{b} \end{vmatrix}, \end{aligned}$$

which is

$$\begin{aligned} &= -\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12})(\mathrm{ab} - \mathrm{h}^2) \\ &\quad + \mathrm{a}\big(\tfrac{1}{2}\mathrm{b}_1^2 - \tfrac{1}{2}\mathrm{b}_2\mathrm{h}_1 + \tfrac{1}{2}\mathrm{a}_2\mathrm{b}_2\big) \\ &\quad + \mathrm{b}\big(-\tfrac{1}{2}\mathrm{a}_2^2 - \tfrac{1}{2}\mathrm{h}_1^2 + \tfrac{1}{2}\mathrm{a}_1\mathrm{h}_2\big) \\ &\quad + \mathrm{h}\big[\tfrac{1}{2}(\mathrm{a}_2\mathrm{b}_2 - \mathrm{a}_1\mathrm{b}_2) + \mathrm{h}_1\mathrm{h}_2 - \mathrm{a}_2\mathrm{h}_2 - \mathrm{b}_1\mathrm{h}_1\big] \\ &= (\mathrm{ab} - \mathrm{h}^2)k, \quad \text{where } k \text{ is the measure of curvature.} \end{aligned}$$

In the same way, we have

$$\begin{aligned} &(23.1.2)(31.1.2) - (33.1.2)(12.1.2) \\ &\quad = \begin{vmatrix} 23.31 & 1.31 & 2.31 \\ 1.23 & 1.1 & 1.2 \\ 2.23 & 1.2 & 2.2 \end{vmatrix} - \begin{vmatrix} 12.33 & 1.33 & 2.33 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 1.2 & 2.2 \end{vmatrix}, \end{aligned}$$

which is

$$= (23.31 - 12.33)\begin{vmatrix} 1.1 & 1.2 \\ 1.2 & 2.2 \end{vmatrix} + \begin{vmatrix} 0 & 1.13 & 2.13 \\ 1.23 & 1.1 & 1.2 \\ 2.23 & 1.2 & 2.2 \end{vmatrix} - \begin{vmatrix} 0 & 1.33 & 2.33 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 1.2 & 2.2 \end{vmatrix},$$

containing $23.31 - 12.33$, which by what precedes is

$$= -\tfrac{1}{2}(\mathrm{g}_{23} + \mathrm{f}_{31} - \mathrm{c}_{12} - \mathrm{h}_{33});$$

the other terms are also known, and the value is

$$\begin{aligned} &= -\tfrac{1}{2}(\mathrm{g}_{23} + \mathrm{f}_{31} - \mathrm{c}_{12} - \mathrm{h}_{33})(\mathrm{ab} - \mathrm{h}^2) \\ &\quad + \mathrm{a}\begin{vmatrix} 0 & \tfrac{1}{2}\mathrm{a}_3 & \tfrac{1}{2}(\mathrm{f}_1 - \mathrm{g}_2 + \mathrm{h}_3) \\ \tfrac{1}{2}(-\mathrm{f}_1 + \mathrm{g}_2 + \mathrm{h}_3) & \mathrm{a} & \mathrm{h} \\ \tfrac{1}{2}\mathrm{b}_3 & \mathrm{h} & \mathrm{b} \end{vmatrix} \\ &\quad - \mathrm{h}\begin{vmatrix} 0 & \tfrac{1}{2}\mathrm{c}_1 & \tfrac{1}{2}\mathrm{c}_2 \\ \tfrac{1}{2}\mathrm{a}_2 & \mathrm{a} & \mathrm{h} \\ \tfrac{1}{2}\mathrm{b}_3 & \mathrm{h} & \mathrm{b} \end{vmatrix}, \end{aligned}$$

or finally this is

$$\begin{aligned} &= -\tfrac{1}{2}(\mathrm{g}_{23} + \mathrm{f}_{31} - \mathrm{c}_{12} - \mathrm{h}_{33})(\mathrm{ab} - \mathrm{h}^2) \\ &\quad + \mathrm{a}\big(\tfrac{1}{2}\mathrm{b}_3\mathrm{f}_1 - \tfrac{1}{2}\mathrm{b}_3\mathrm{g}_2 - \tfrac{1}{2}\mathrm{b}_3\mathrm{h}_3 + \tfrac{1}{2}\mathrm{a}_3\mathrm{b}_2 - \mathrm{b}_3\mathrm{h}_3\big) \\ &\quad + \mathrm{b}\big(\tfrac{1}{2}\mathrm{a}_3\mathrm{g}_2 + \dots\big) \\ &\quad + \mathrm{h}\big(\dots\big). \end{aligned}$$

The remaining four of the six functions can of course be obtained from the two just found by a cyclical interchange of the letters and suffix-numbers 1, 2, 3, and it is not worth while to write down the values.

The two values of $(11.1.2)(22.1.2) - (12.1.2)^2$ are

$$\begin{aligned} &-\tfrac{1}{2}(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12})(\mathrm{ab} - \mathrm{h}^2) + \mathrm{a}(\tfrac{1}{2}\mathrm{b}_1^2 - \tfrac{1}{2}\mathrm{b}_2\mathrm{h}_1 + \tfrac{1}{2}\mathrm{a}_2\mathrm{b}_2) \\ &\quad + \mathrm{b}(-\tfrac{1}{2}\mathrm{a}_2^2 - \tfrac{1}{2}\mathrm{h}_1^2 + \tfrac{1}{2}\mathrm{a}_1\mathrm{h}_2) \\ &\quad + \mathrm{h}\big[\tfrac{1}{2}(\mathrm{a}_2\mathrm{b}_2 - \mathrm{a}_1\mathrm{b}_2) + \mathrm{h}_1\mathrm{h}_2 - \mathrm{a}_2\mathrm{h}_2 - \mathrm{b}_1\mathrm{h}_1\big], \end{aligned}$$

and

$$\{G \cdot \tfrac{1}{2}\mathrm{a}_1 + F(\mathrm{h}_1 - \tfrac{1}{2}\mathrm{a}_2) + C(\mathrm{g}_1 - \tfrac{1}{2}\mathrm{a}_3)\} \cdot \{G(\mathrm{h}_2 - \tfrac{1}{2}\mathrm{b}_1) + F \cdot \tfrac{1}{2}\mathrm{b}_2 + C(\mathrm{f}_2 - \tfrac{1}{2}\mathrm{b}_3)\},$$

where, in the first value, for a, b, h, $\mathrm{ab} - \mathrm{h}^2$ we must write $L^{-2}(BC - F^2)$, $L^{-2}(CA - G^2)$, $L^{-2}(FG - CH)$, and $L^{-2}C^2$; making this change, multiplying by 4 and equating, we obtain

$$\begin{aligned} &-2L^2C(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}) \\ &\quad + (BC - F^2)(\mathrm{b}_1^2 - 2\mathrm{b}_2\mathrm{h}_1 + \mathrm{a}_2\mathrm{b}_2) \\ &\quad + (CA - G^2)(\mathrm{a}_2^2 - 2\mathrm{a}_1\mathrm{h}_2 + \mathrm{a}_1\mathrm{b}_1) \\ &\quad + (FG - CH)(\mathrm{a}_1\mathrm{b}_2 - \mathrm{a}_2\mathrm{b}_1 + 4\mathrm{h}_1\mathrm{h}_2 - 2\mathrm{b}_1\mathrm{h}_1 - 2\mathrm{a}_2\mathrm{h}_2) \\ &\quad - \{G \cdot \mathrm{a}_1 + F(2\mathrm{h}_1 - \mathrm{a}_2) + C(2\mathrm{g}_1 - \mathrm{a}_3)\}\{G(2\mathrm{h}_2 - \mathrm{b}_1) + F\mathrm{b}_2 + C(2\mathrm{f}_2 - \mathrm{b}_3)\} \\ &\quad + \{G\mathrm{b}_1 + F\mathrm{b}_2 + C(\mathrm{f}_1 + \mathrm{g}_2 - \mathrm{h}_3)\}^2 \;=\; 0. \end{aligned}$$

C. XII. 2

Developing the fourth and fifth lines, it appears that in this expression the coefficients of F^2, G^2 and FG each of them vanish; the whole equation is thus divisible by C, and omitting this factor throughout, the equation becomes

$$\begin{aligned}
0 = &-2L(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12})\\
&+ A(\mathrm{a}_2^2 - 2\mathrm{a}_2\mathrm{h}_1 + \mathrm{a}_1\mathrm{b}_1)\\
&+ B(\mathrm{b}_1^2 - 2\mathrm{b}_1\mathrm{h}_2 + \mathrm{a}_2\mathrm{b}_2)\\
&+ C\big\{ - \mathrm{a}_3\mathrm{b}_3 + 2\mathrm{a}_2\mathrm{f}_3 + 2\mathrm{b}_1\mathrm{g}_3 - 4\mathrm{f}_3\mathrm{g}_3 + (\mathrm{f}_1 + \mathrm{g}_2 - \mathrm{h}_3)^2\big\}\\
&+ F\big\{(\mathrm{a}_3\mathrm{b}_2 - \mathrm{a}_2\mathrm{b}_3) + 2(\mathrm{b}_1\mathrm{g}_3 - \mathrm{b}_2\mathrm{g}_1) + 2(\mathrm{b}_1\mathrm{h}_3 - \mathrm{b}_3\mathrm{h}_1) + 2(\mathrm{a}_2\mathrm{f}_2 + \mathrm{b}_1\mathrm{f}_2 - 2\mathrm{h}_1\mathrm{f}_2)\big\}\\
&+ G\big\{\mathrm{a}_3\mathrm{b}_2 - \mathrm{a}_2\mathrm{b}_3 + 2(\mathrm{a}_2\mathrm{f}_3 - \mathrm{a}_3\mathrm{f}_2) + 2(\mathrm{a}_2\mathrm{h}_3 - \mathrm{a}_3\mathrm{h}_2) + 2(\mathrm{a}_2\mathrm{g}_2 + \mathrm{b}_1\mathrm{g}_1 - 2\mathrm{g}_1\mathrm{h}_2)\big\}\\
&+ H\big\{\ 2(\mathrm{a}_2\mathrm{h}_2 + \mathrm{b}_1\mathrm{h}_1 - 2\mathrm{h}_1\mathrm{h}_2)\big\};
\end{aligned}$$

this is substantially the required equation ($\mathfrak{C}$), but the form of it may be greatly simplified.

Forming the identity,

$$\begin{aligned}
0 &= 2L[(\mathrm{a}_2 - \mathrm{h}_1)L_1 + (\mathrm{b}_1 - \mathrm{h}_2)L_2]\\
&= -A(\mathrm{a}_1\mathrm{h}_2 - \mathrm{a}_2\mathrm{h}_2 - \mathrm{a}_1\mathrm{h}_1 + \mathrm{a}_2^2)\\
&\quad - B(\mathrm{b}_1^2 - \mathrm{b}_1\mathrm{h}_2 - \mathrm{b}_2\mathrm{h}_1 + \mathrm{a}_2\mathrm{b}_2)\\
&\quad - C(\mathrm{b}_1\mathrm{c}_2 + \mathrm{a}_2\mathrm{c}_1 - \mathrm{c}_1\mathrm{h}_2 - \mathrm{c}_2\mathrm{h}_1)\\
&\quad - 2F(\mathrm{b}_1\mathrm{f}_2 + \mathrm{a}_2\mathrm{f}_1 - \mathrm{f}_1\mathrm{h}_2 - \mathrm{f}_2\mathrm{h}_1)\\
&\quad - 2G(\mathrm{b}_1\mathrm{g}_2 + \mathrm{a}_2\mathrm{g}_1 - \mathrm{g}_1\mathrm{h}_2 - \mathrm{g}_2\mathrm{h}_1)\\
&\quad - 2H(\mathrm{a}_2\mathrm{h}_2 + \mathrm{b}_1\mathrm{h}_1 - 2\mathrm{h}_1\mathrm{h}_2),
\end{aligned}$$

we add hereto the last preceding equation; the coefficients of A, B, F, G, H thus assume new and simple forms, but the coefficient of C requires a further transformation.

Assume

$$\begin{aligned}
\Omega = &(\mathrm{b}_1\mathrm{c}_1 - \mathrm{c}_2\mathrm{h}_1) + (\mathrm{c}_2\mathrm{a}_2 - \mathrm{c}_1\mathrm{g}_2) + (\mathrm{a}_3\mathrm{b}_3 - \mathrm{h}_3^2)\\
&+ (\mathrm{g}_2\mathrm{h}_3 + \mathrm{g}_3\mathrm{h}_2 - \mathrm{a}_2\mathrm{f}_3 - \mathrm{a}_3\mathrm{f}_2) + (\mathrm{h}_3\mathrm{f}_1 + \mathrm{h}_1\mathrm{f}_3 - \mathrm{b}_1\mathrm{g}_3 - \mathrm{b}_3\mathrm{g}_1) + (\mathrm{f}_1\mathrm{g}_2 + \mathrm{f}_2\mathrm{g}_1 - \mathrm{c}_2\mathrm{h}_1 - \mathrm{c}_1\mathrm{h}_2);
\end{aligned}$$

then, if we add to the equation C multiplied by this value and subtract $C\Omega$, the coefficient of C takes its proper form, and the equation is

$$\begin{aligned}
0 = &- 2L(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12})\\
&+ 2L[(\mathrm{b}_1 - \mathrm{h}_2)L_2 + (\mathrm{a}_2 - \mathrm{h}_1)L_1] - C\Omega\\
&+ A\{\ (\mathrm{a}_1\mathrm{h}_2 - \mathrm{a}_2\mathrm{h}_1)\ \}\\
&+ B\{\ (\mathrm{b}_1\mathrm{h}_2 - \mathrm{b}_2\mathrm{h}_1)\ \}\\
&+ C\{-(\mathrm{a}_2\mathrm{f}_2 - \mathrm{a}_1\mathrm{f}_1) + (\mathrm{b}_1\mathrm{g}_1 - \mathrm{b}_2\mathrm{g}_2) - (\mathrm{g}_2\mathrm{h}_1 - \mathrm{g}_1\mathrm{h}_2) - (\mathrm{h}_1\mathrm{f}_2 - \mathrm{h}_2\mathrm{f}_1) + 3(\mathrm{f}_1\mathrm{g}_2 - \mathrm{f}_2\mathrm{g}_1)\}\\
&+ F\{-(\mathrm{a}_2\mathrm{b}_2 - \mathrm{a}_2\mathrm{b}_3) + 2(\mathrm{b}_1\mathrm{g}_3 - \mathrm{b}_3\mathrm{g}_1) + 2(\mathrm{b}_1\mathrm{h}_3 - \mathrm{b}_3\mathrm{h}_1) - 2(\mathrm{h}_1\mathrm{f}_3 - \mathrm{h}_3\mathrm{f}_1)\}\\
&+ G\{-(\mathrm{a}_2\mathrm{b}_3 - \mathrm{a}_3\mathrm{b}_2) - 2(\mathrm{a}_2\mathrm{f}_3 - \mathrm{a}_3\mathrm{f}_2) - 2(\mathrm{a}_2\mathrm{h}_3 - \mathrm{a}_3\mathrm{h}_2) - 2(\mathrm{g}_1\mathrm{h}_3 - \mathrm{g}_3\mathrm{h}_1)\}\\
&+ H\{\ \cdots\ \}\\
= &\ 0;
\end{aligned}$$

where, as a verification, observe that the letters (a, b), (f, g) may be interchanged if at the same time we interchange the suffixes 1 and 2.

The two values of $(23.1.2)(31.1.2) - (33.1.2)(12.1.2)$ are

$$\begin{aligned}
&-\tfrac{1}{2}(\mathrm{g}_{23}+\mathrm{f}_{31}-\mathrm{c}_{12}-\mathrm{h}_{33})(\mathrm{ab}-\mathrm{h}^2)\\
&\quad+\mathrm{a}\big(\tfrac{1}{2}\mathrm{b}_1\mathrm{f}_3-\tfrac{1}{2}\mathrm{b}_3\mathrm{f}_1-\tfrac{1}{2}\mathrm{b}_1\mathrm{c}_2+\tfrac{1}{2}\mathrm{b}_2\mathrm{g}_3-\mathrm{b}_3\mathrm{h}_3\big)\\
&\quad+\mathrm{b}\big(\tfrac{1}{2}\mathrm{a}_3\mathrm{g}_2+\ldots-\mathrm{a}_3\mathrm{h}_3\big)+\mathrm{h}\big[\ldots-\tfrac{1}{2}(\mathrm{f}_1+\mathrm{g}_2)^2\big],
\end{aligned}$$

and

$$\{G\cdot\tfrac{1}{2}\mathrm{a}_3+F(\mathrm{f}_2-\tfrac{1}{2}\mathrm{c}_1)+C\cdot\tfrac{1}{2}\mathrm{c}_3\}\cdot\{G\cdot\tfrac{1}{2}\mathrm{a}_2+F\cdot\tfrac{1}{2}\mathrm{b}_3+C\cdot\tfrac{1}{2}(\mathrm{f}_1+\mathrm{g}_2-\mathrm{h}_3)\},$$

and here for a, b, h, $\mathrm{ab}-\mathrm{h}^2$ substituting their values, multiplying by 4, and equating, we have

$$\begin{aligned}
&-2L^2C(\mathrm{g}_{23}+\mathrm{f}_{31}-\mathrm{c}_{12}-\mathrm{h}_{33})\\
&\quad+(BC-F^2)(2\mathrm{b}_1\mathrm{f}_3-\mathrm{b}_3\mathrm{f}_1-\mathrm{b}_1\mathrm{c}_2+\mathrm{b}_2\mathrm{g}_3-\mathrm{b}_3\mathrm{h}_3)\\
&\quad+(CA-G^2)(2\mathrm{a}_2\mathrm{g}_3-\mathrm{a}_3\mathrm{g}_2-\mathrm{a}_2\mathrm{c}_1+\mathrm{a}_1\mathrm{f}_3-\mathrm{a}_3\mathrm{h}_3)\\
&\quad+(FG-CH)(\mathrm{a}_3\mathrm{b}_3+\mathrm{b}_1\mathrm{c}_1+\mathrm{a}_2\mathrm{c}_2-2\mathrm{b}_1\mathrm{g}_3-2\mathrm{a}_2\mathrm{f}_3+\mathrm{h}_3^2-\mathrm{f}_2\mathrm{g}_1+2\mathrm{f}_1\mathrm{g}_2-\mathrm{g}_1^2\mathrm{a}_2^2)\\
&\quad-\{G(-\mathrm{f}_1+\mathrm{g}_2+\mathrm{h}_3)+F\mathrm{b}_3+C\mathrm{c}_2\}\{G\mathrm{a}_3+F(\mathrm{f}_2-\mathrm{g}_1+\mathrm{h}_2)+C\mathrm{c}_1\}\\
&\quad+\{G(2\mathrm{g}_3-\mathrm{c}_2)+F(2\mathrm{f}_3-\mathrm{c}_2)+C\mathrm{c}_3\}\{G\mathrm{a}_2+F\mathrm{b}_2+C(\mathrm{f}_1+\mathrm{g}_2-\mathrm{h}_3)\}=0.
\end{aligned}$$

Here again the terms in F^2, FG, G^2 all vanish, hence the whole equation divides by C; throwing this factor out, we have

$$\begin{aligned}
0=\ &-2L(\mathrm{g}_{23}+\mathrm{f}_{31}-\mathrm{c}_{12}-\mathrm{h}_{33})\\
&+A(\ 2\mathrm{a}_2\mathrm{g}_3-\mathrm{a}_3\mathrm{g}_2-\mathrm{a}_2\mathrm{c}_1+\mathrm{a}_1\mathrm{f}_3-\mathrm{a}_3\mathrm{h}_3\ )\\
&+B(\ 2\mathrm{b}_1\mathrm{f}_3-\mathrm{b}_3\mathrm{f}_1-\mathrm{b}_1\mathrm{c}_2+\mathrm{b}_2\mathrm{g}_3-\mathrm{b}_3\mathrm{h}_3\ )\\
&+C(\ -\mathrm{c}_1\mathrm{c}_2+\mathrm{c}_3\mathrm{f}_1+\mathrm{c}_3\mathrm{g}_2-\mathrm{c}_3\mathrm{h}_3\ )\\
&+F\{\ -(\mathrm{b}_3\mathrm{c}_1-\mathrm{b}_1\mathrm{c}_3)-2\mathrm{c}_2\mathrm{f}_1+2\mathrm{f}_1\mathrm{f}_3+2\mathrm{g}_2\mathrm{f}_3-2\mathrm{f}_3\mathrm{h}_3\ \}\\
&+G(\ \mathrm{a}_2\mathrm{c}_3-\mathrm{a}_3\mathrm{c}_2-2\mathrm{c}_1\mathrm{g}_3+2\mathrm{f}_2\mathrm{g}_3+2\mathrm{g}_2\mathrm{g}_3-2\mathrm{g}_3\mathrm{h}_3\ )\\
&+H(\ -\mathrm{a}_3\mathrm{b}_3-\mathrm{b}_1\mathrm{c}_1-\mathrm{a}_2\mathrm{c}_2+2\mathrm{b}_1\mathrm{g}_3+2\mathrm{a}_2\mathrm{f}_3-\mathrm{h}_3^2+\mathrm{f}_2\mathrm{g}_1-2\mathrm{f}_1\mathrm{g}_2+\mathrm{g}_1^2\mathrm{a}_2^2\ )=0,
\end{aligned}$$

which is substantially the required equation ($\mathfrak{f}$); but the form has to be altered. First, multiply by 2, then forming the expression

$$\begin{aligned}
&2L[(\mathrm{f}_3-\mathrm{c}_2)L_1+(\mathrm{g}_3-\mathrm{c}_1)L_2+(\mathrm{f}_1+\mathrm{g}_2-2\mathrm{h}_3)L_3]\\
&\qquad=(\mathrm{f}_3-\mathrm{c}_2)(A\mathrm{a}_1+B\mathrm{b}_1+C\mathrm{c}_1+2F\mathrm{f}_1+\ldots)\\
&\qquad\quad+(\mathrm{g}_3-\mathrm{c}_1)(A\mathrm{a}_2+B\mathrm{b}_2+C\mathrm{c}_2+2F\mathrm{f}_2+\ldots)\\
&\qquad\quad+(\mathrm{f}_1+\mathrm{g}_2-2\mathrm{h}_3)(A\mathrm{a}_3+B\mathrm{b}_3+C\mathrm{c}_3+2F\mathrm{f}_3+2G\mathrm{g}_3+\ldots),
\end{aligned}$$

2—2

this is

$$\begin{aligned}
&= A(-a_1c_2 - a_2c_1 + a_1f_1 + a_2f_1 + a_2g_2 + a_3g_3 - 2a_3h_3)\\
&\quad + B(-b_1c_2 - b_2c_1 + b_1f_1 + b_1f_3 + b_2g_3 + b_3g_2 - 2b_3h_3)\\
&\quad + C(-2c_1c_2 + c_1f_1 + c_3f_1 + c_2g_2 + c_3g_2 - 2c_3h_3)\\
&\quad + 2F(-c_1f_1 - c_2f_2 + 2f_1f_3 + f_2g_3 + f_3g_2 - 2f_3h_3)\\
&\quad + 2G(-c_1g_2 - c_2g_1 + f_1g_2 + f_2g_1 + 2g_2g_3 - 2g_3h_3)\\
&\quad + 2H(-c_1h_1 - c_2h_2 + g_1h_2 + g_2h_1 + h_1f_2 + h_2f_1 - 2h_3^2);
\end{aligned}$$

then adding this expression

$$(f_3 - c_2)L_1 + (g_3 - c_1)L_2 + (f_1 + g_2 - 2h_3)L_3$$

and subtracting its foregoing value, we have an equation with new coefficients of A, B, C, F, G, H, all of which, except that of H, are in the proper form, and for the coefficient of H, recurring to the foregoing value of Ω, we must add to the equation $2H$ multiplied by the value of Ω and subtract $2H\Omega$. The final result is

$$\begin{aligned}
0 = &- 4L(g_{23} + f_{31} - c_{12} - h_{33})\\
&+ 2L[(f_3 - c_2)L_1 + (g_3 - c_1)L_2 + (f_1 + g_2 - 2h_3)L_3] - 2H\Omega\\
&+ A\{\ 3(a_2g_3 - a_3g_2) - (c_1a_2 - c_2a_1) + (a_1f_3 - a_3f_1)\ \}\\
&+ B\{\ -3(b_3f_1 - b_1f_3) - (b_1c_2 - b_2c_1) - (b_3g_2 - b_2g_3)\ \}\\
&+ C\{\ (c_2f_1 - c_1f_2) - (c_1g_2 - c_2g_1)\ \}\\
&+ 2F\{\ -(b_3c_1 - b_1c_3) + (c_1f_2 - c_2f_1) - (f_2g_3 - f_3g_2)\ \}\\
&+ 2G\{\ -(c_3a_2 - c_2a_3) - (c_1g_2 - c_2g_1) - (f_1g_3 - f_3g_1)\ \}\\
&+ 2H\{\ -(b_3g_1 - b_1g_3) + (a_2f_3 - a_3f_2)\ \} = 0,
\end{aligned}$$

which, it will be observed, remains unaltered by the interchange of a and b, f and g, A and B, F and G, the suffix numbers 1 and 2 being at the same time interchanged.

We have thus the required six equations, in which

$$\mathfrak{K} = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

$$\begin{aligned}
\Omega = &\ b_1c_1 - c_2h_1 + c_2a_2 - c_1g_2 + a_3b_3 - h_3^2\\
&+ (g_2h_3 + g_3h_2 - a_2f_3 - a_3f_2) + (h_3f_1 + h_1f_3 - b_1g_3 - b_3g_1) + (f_1g_2 + f_2g_1 - c_2h_1 - c_1h_2),
\end{aligned}$$

and where I write also, for shortness, ab|2 for $a_1b_2 - a_2b_1$, &c.; viz. the equations are

$$\begin{aligned}
0 = &- 2L(b_{33} + c_{22} - 2f_{23})\\
&+ 2L[(c_3 - f_2)L_2 + (b_2 - f_3)L_3] - A\Omega\\
&+ A\{\ -\mathrm{bg}|31 + \mathrm{ch}|2 + 3\cdot\mathrm{gh}|23 - \mathrm{hf}|31 - \mathrm{fg}|12\ \}\\
&+ B\{\ -\mathrm{bf}|23\ \}\\
&+ C\{\ +\mathrm{cf}|23\ \}\\
&+ F\{\ -\mathrm{bc}|23\ \}\\
&+ G\{\ -\mathrm{bc}|31 + 2\cdot\mathrm{cf}|12 + 2\cdot\mathrm{ch}|23 - 2\cdot\mathrm{fg}|23\ \}\\
&+ H\{\ -2\cdot\mathrm{bg}|23 - 2\cdot\mathrm{hf}|23\ \} \qquad (\mathfrak{A}).
\end{aligned}$$

$$
\begin{aligned}
0 = &- 2L(\mathrm{c}_{11} + \mathrm{a}_{33} - 2\mathrm{g}_{31}) \\
&+ 2L[(\mathrm{c}_3 - \mathrm{g}_1)L_1 + (\mathrm{a}_1 - \mathrm{g}_3)L_3] - B\Omega \\
&+ A\{\ \mathrm{ag}|31\ \} \\
&+ B\{\ \mathrm{af}|23 - \mathrm{ch}|12 - \mathrm{gh}|23 + 3\cdot\mathrm{hf}|31 - \mathrm{fg}|31\ \} \\
&+ C\{\ -\mathrm{cg}|31\ \} \\
&+ F\{\ -\mathrm{ca}|23 - 2\cdot\mathrm{cg}|12 - 2\cdot\mathrm{ch}|31 - 2\cdot\mathrm{fg}|31\ \} \\
&+ G\{\ -\mathrm{ca}|31 - 2\cdot\mathrm{gh}|31\ \} \\
&+ H\{\ \ldots\ \} \qquad (\mathfrak{B}).
\end{aligned}
$$

$$
\begin{aligned}
0 = &- 2L(\mathrm{a}_{22} + \mathrm{b}_{11} - 2\mathrm{h}_{12}) \\
&+ 2L[(\mathrm{b}_1 - \mathrm{h}_2)L_2 + (\mathrm{a}_2 - \mathrm{h}_1)L_1] - C\Omega \\
&+ A\{\ -\mathrm{ah}|12\ \} \\
&+ B\{\ +\mathrm{bh}|12\ \} \\
&+ C\{\ -\mathrm{af}|23 + \mathrm{bg}|31 - \mathrm{gh}|23 - \mathrm{hf}|31 + 3\cdot\mathrm{fg}|12\ \} \\
&+ F\{\ -\mathrm{ab}|23 + 2\cdot\mathrm{bg}|12 + 2\cdot\mathrm{bh}|31 - 2\cdot\mathrm{hf}|12\ \} \\
&+ G\{\ -\mathrm{ab}|31 - 2\cdot\mathrm{af}|12 + 2\cdot\mathrm{ah}|23 - 2\cdot\mathrm{gh}|12\ \} \\
&+ H\{\ -\mathrm{ab}|12\ \} \qquad (\mathfrak{C}).
\end{aligned}
$$

$$
\begin{aligned}
0 = &- 4L(\mathrm{g}_{12} + \mathrm{h}_{31} - \mathrm{a}_{23} - \mathrm{f}_{11}) \\
&+ 2L[(\mathrm{g}_2 + \mathrm{h}_3 - 2\mathrm{f}_1)L_1 + (\mathrm{g}_3 - \mathrm{a}_1)L_2 + (\mathrm{h}_1 - \mathrm{a}_1)L_3] - 2F\Omega \\
&+ A\{\ -\mathrm{ag}|12 - \mathrm{ah}|31\ \} \\
&+ B\{\ -\mathrm{ab}|23 + \mathrm{bg}|12 + 3\cdot\mathrm{bh}|31\ \} \\
&+ C\{\ -\mathrm{ca}|23 - 3\cdot\mathrm{cg}|12 - \mathrm{ch}|31\ \} \\
&+ 2F\{\ \ldots + \mathrm{bg}|31 - \mathrm{ch}|12\ \} \\
&+ 2G\{\ -\mathrm{ca}|12 + \mathrm{ag}|23 - \mathrm{gh}|31\ \} \\
&+ 2H\{\ -\mathrm{ab}|31 - \mathrm{ah}|23 - \mathrm{gh}|12\ \} \qquad (\mathfrak{f}).
\end{aligned}
$$

$$
\begin{aligned}
0 = &- 4L(\mathrm{h}_{12} + \mathrm{f}_{31} - \mathrm{b}_{23} - \mathrm{g}_{22}) \\
&+ 2L[(\mathrm{f}_2 - \mathrm{b}_3)L_1 + (\mathrm{h}_1 + \mathrm{f}_3 - 2\mathrm{g}_2)L_2 + (\mathrm{h}_2 - \mathrm{b}_1)L_3] - 2G\Omega \\
&+ A\{\ -\mathrm{ab}|31 - 3\cdot\mathrm{ah}|23 - \mathrm{af}|12\ \} \\
&+ B\{\ \mathrm{bh}|23 - \mathrm{bf}|12\ \} \\
&+ C\{\ -\mathrm{bc}|31 + \mathrm{ch}|23 + 3\cdot\mathrm{cf}|12\ \} \\
&+ 2F\{\ -\mathrm{bc}|12 - \mathrm{bf}|31 - \mathrm{hf}|23\ \} \\
&+ 2G\{\ \ldots + \mathrm{ch}|12 - \mathrm{af}|23\ \} \\
&+ 2H\{\ -\mathrm{ab}|23 + \mathrm{bh}|31 - \mathrm{hf}|12\ \} \qquad (\mathfrak{g}).
\end{aligned}
$$

$$
\begin{aligned}
0 = & -4L(\mathrm{f}_{12}+\mathrm{g}_{31}-\mathrm{c}_{23}-\mathrm{h}_{33}) \\
& +2L[(\mathrm{f}_3-\mathrm{c}_2)L_1+(\mathrm{g}_3-\mathrm{c}_1)L_2+(\mathrm{f}_1+\mathrm{g}_2-2\mathrm{h}_3)L_3]-2H\Omega \\
& +A\{\ -\mathrm{ca}|12+\mathrm{af}|31+3\cdot\mathrm{ag}|23\ \} \\
& +B\{\ -\mathrm{bc}|12-3\cdot\mathrm{bf}|31-\mathrm{bg}|23\ \} \\
& +C\{\ \mathrm{cf}|31-\mathrm{cg}|23\ \} \\
& +F\{\ -\mathrm{bc}|31+\mathrm{cf}|12-\mathrm{fg}|23\ \} \\
& +G\{\ -\mathrm{ca}|23-\mathrm{cg}|12-\mathrm{fg}|31\ \} \\
& +H\{\ \ldots+\mathrm{af}|23-\mathrm{bg}|31\ \} \qquad (\mathfrak{h}).
\end{aligned}
$$

It would be possible in these equations to introduce the symbols $AB|12$, &c., in place of ab$|$12, &c., and then writing $A = B = C = 1$, all these symbols other than those where the letters are GH, HF or FG would vanish, and we should obtain Mr Warren's six equations for normal coordinates. But in the general case it would seem that there is not any advantage in the introduction of the new symbols $AB|12$, &c., and I retain by preference the equations in the form in which I have given them.

To the foregoing may be joined a symmetrical equation obtained (as by Mr Warren) by multiplying the several equations by a, b, c, f, g, h respectively, and adding; the result is in the first instance obtained in the form

$$\mathfrak{X}-\mathfrak{Y}+\mathfrak{Z}=0,$$

where

$$
\begin{aligned}
\mathfrak{X} = & \ \mathrm{a}(\mathrm{b}_{33}+\mathrm{c}_{22}-2\mathrm{f}_{23}) \\
& +\mathrm{b}(\mathrm{c}_{11}+\mathrm{a}_{33}-2\mathrm{g}_{31}) \\
& +\mathrm{c}(\mathrm{a}_{22}+\mathrm{b}_{11}-2\mathrm{h}_{12}) \\
& +2\mathrm{f}(\mathrm{g}_{12}+\mathrm{h}_{31}-\mathrm{a}_{23}-\mathrm{f}_{11}) \\
& +2\mathrm{g}(\mathrm{h}_{12}+\mathrm{f}_{31}-\mathrm{b}_{23}-\mathrm{g}_{22}) \\
& +2\mathrm{h}(\mathrm{f}_{12}+\mathrm{g}_{31}-\mathrm{c}_{23}-\mathrm{h}_{33}).
\end{aligned}
$$

For Ψ, collecting the terms which contain L_1, L_2, L_3 respectively, and attending to the values of (A, B, C, F, G, H),

$$= (\mathrm{bc}-\mathrm{f}^2,\ \mathrm{ca}-\mathrm{g}^2,\ \mathrm{ab}-\mathrm{h}^2,\ \mathrm{gh}-\mathrm{af},\ \mathrm{hf}-\mathrm{bg},\ \mathrm{fg}-\mathrm{ch}),$$

this is easily reduced to the form

$$\Psi = (A_1+H_2+G_3)L_1+(H_1+B_2+F_3)L_2+(G_1+F_2+C_3)L_3.$$

The term in Ω is

$$= -(A\mathrm{a}+B\mathrm{b}+C\mathrm{c}+2F\mathrm{f}+2G\mathrm{g}+2H\mathrm{h})\Omega = -3L^2\Omega, \text{ as above.}$$

For the calculation of $\square$, collecting the terms, we have $\square =$

23	A	B	C	F	G	H
bc				$-a$		
ca			$-f$	$-b-2h$		
ab		$-f$	$-c$			$-2g$
af		$b-c$			$-2g$	$2h$
bf		$-a$				
cf			a			
ag	$3h$				$2f$	$2b$
bg		$-h$				$-2a$
cg			$-h$			
ah	$-3g$			$-2c-2f$		
bh		g				
ch			g		$2a$	
gh	$-3a-b-c$					
hf				$-2g$		$-2a$
fg				$-2h-2a$		

31	A	B	C	F	G	H
bc			$-g$	$-2h$	$-a$	
ca					$-b$	
ab	$-g$				$-c$	$-2f$
af	h					$2b$
bf		$-3h$		$-2g$		$-2a$
cf			h			
ag	b					
bg	$-a$		$+c$	$2f$		$-2h$
cg			$-b$			
ah	$-f$					
bh		$3f$		$2c$		$2g$
ch			$-f-2b$			
gh				$-2f$		$-2b$
hf	$-a$	$3b-c$				
fg				$-2b-2h$		

12	A	B	C	F	G	H
bc		$-h$		$-2g$		$-a$
ca	$-h$				$-2f$	$-b$
ab						$-c$
af	$-g$				$-2c$	
bf		$-g$				
cf			$3g$	$2h$	$2a$	
ag	f					
bg		f		$2c$		
cg			$-3f-2b$	$-2h$		
ah	$-c$					
bh		c				
ch	$a-b$			$-2f+2g$		
gh					$-2c$	$-2f$
hf				$-2c$		$-2g$
fg	$-a-b+3c$					

Some however of the coefficients require reduction; for instance, that of bc31 is

$$= -(\mathrm{a}, \mathrm{h}, \mathrm{g}\rangle G, F, C) - \mathrm{h}F = -\mathrm{h}F;$$

and so $-\mathrm{h}B - 2\mathrm{g}F - \mathrm{a}H$ is

$$= -(\mathrm{a}, \mathrm{h}, \mathrm{g}\rangle H, B, F) - \mathrm{g}F = -\mathrm{g}F.$$

After these reductions, the value is found to be or, arranging the terms in a different order, this may be written

$$\square =$$

$$\begin{aligned}
&A\{\ \mathrm{a}(3\cdot \mathrm{gh}|23-\mathrm{hf}|31-\mathrm{fg}|12)+\mathrm{b}\cdot \mathrm{ag}|31-\mathrm{c}\cdot \mathrm{ah}|12-\mathrm{c}\cdot \mathrm{ah}|12\\
&\qquad +\mathrm{f}(\mathrm{ag}|12-\mathrm{ah}|31)+\mathrm{g}(2\cdot \mathrm{gh}|12-\mathrm{ah}|23)+\mathrm{h}(2\cdot \mathrm{gh}|31+\mathrm{ag}|23)\ \}\\
&+B\{\ -\mathrm{a}\cdot \mathrm{bf}|23+\mathrm{b}(-\mathrm{gh}|23+3\cdot \mathrm{hf}|31-\mathrm{fg}|12)+\mathrm{c}\cdot \mathrm{bh}|12\\
&\qquad +\mathrm{f}(2\cdot \mathrm{hf}|12+\mathrm{bh}|31)+\mathrm{g}(\mathrm{bh}|23-\mathrm{bf}|12)+\mathrm{h}(2\cdot \mathrm{hf}|23-\mathrm{bf}|31)\ \}\\
&+C\{\ \mathrm{a}\cdot \mathrm{cf}|23-\mathrm{b}\cdot \mathrm{cg}|31+\mathrm{c}(-\mathrm{gh}|23-\mathrm{hf}|31+3\cdot \mathrm{fg}|12)\\
&\qquad +\mathrm{f}(2\cdot \mathrm{fg}|31-\mathrm{cg}|12)+\mathrm{g}(2\cdot \mathrm{fg}|23+\mathrm{cf}|12)+\mathrm{h}(\mathrm{cf}|31-\mathrm{cg}|23)\ \}\\
&+F\{\ -\mathrm{a}\cdot \mathrm{bc}|23-\mathrm{b}\cdot \mathrm{ch}|31+\mathrm{c}\cdot \mathrm{bg}|12\\
&\qquad +\mathrm{f}(\mathrm{bg}|31-\mathrm{ch}|12)+\mathrm{g}(\mathrm{bg}|23-\mathrm{bc}|12)-\mathrm{h}(\mathrm{ch}|23+\mathrm{bc}|31)\ \}\\
&+G\{\ +\mathrm{a}\cdot \mathrm{ch}|23-\mathrm{b}\cdot \mathrm{ca}|31-\mathrm{c}\cdot \mathrm{af}|12\\
&\qquad -\mathrm{f}(\mathrm{af}|31+\mathrm{ca}|12)+\mathrm{g}(\mathrm{ch}|12-\mathrm{af}|23)+\mathrm{h}(\mathrm{ch}|31-\mathrm{ca}|23)\ \}\\
&+H\{\ -\mathrm{a}\cdot \mathrm{bg}|23+\mathrm{b}\cdot \mathrm{af}|31-\mathrm{c}\cdot \mathrm{ab}|12\\
&\qquad +\mathrm{f}(\mathrm{af}|12-\mathrm{ab}|31)-\mathrm{g}(\mathrm{bg}|12+\mathrm{ab}|23)+\mathrm{h}(\mathrm{af}|23-\mathrm{bg}|31)\ \}.
\end{aligned}$$

Attending to the values of A, B, C, F, G, H, we have

$$\begin{aligned}
A_1+B_2+C_3+2F_4+2G_5+2H_6 &= \mathrm{bc}_{11}+\mathrm{cb}_{11}-2\mathrm{ff}_{11}+2(\mathrm{b}_1\mathrm{c}_1-\mathrm{f}_1^2)\\
&\quad +\mathrm{ca}_{22}+\mathrm{ac}_{22}-2\mathrm{gg}_{22}+2(\mathrm{c}_2\mathrm{a}_2-\mathrm{g}_2^2)\\
&\quad +\mathrm{ab}_{33}+\mathrm{ba}_{33}-2\mathrm{hh}_{33}+2(\mathrm{a}_3\mathrm{b}_3-\mathrm{h}_3^2)\\
&\quad +2\{\mathrm{gh}_{23}+\mathrm{hg}_{23}-\mathrm{af}_{23}-\mathrm{fa}_{23}+\mathrm{g}_2\mathrm{h}_3+\mathrm{g}_3\mathrm{h}_2-\mathrm{a}_2\mathrm{f}_3-\mathrm{a}_3\mathrm{f}_2\}\\
&\quad +2\{\mathrm{hf}_{31}+\mathrm{fh}_{31}-\mathrm{bg}_{31}-\mathrm{gb}_{31}+\mathrm{h}_3\mathrm{f}_1+\mathrm{h}_1\mathrm{f}_3-\mathrm{b}_3\mathrm{g}_1-\mathrm{b}_1\mathrm{g}_3\}\\
&\quad +2\{\mathrm{fg}_{12}+\mathrm{gf}_{12}-\mathrm{ch}_{12}-\mathrm{hc}_{12}+\mathrm{f}_1\mathrm{g}_2+\mathrm{f}_2\mathrm{g}_1-\mathrm{c}_1\mathrm{h}_2-\mathrm{c}_2\mathrm{h}_1\},
\end{aligned}$$

which is, in fact, $=2\mathfrak{X}+2\Omega$.

Hence the foregoing equation $(\mathfrak{M})$ may also be written

$$-L^2\,(A_{11}+B_{22}+C_{33}+2F_{23}+2G_{31}+2H_{12})+2L\Psi-L^2\Omega+\square=0,$$

where Ψ, Ω, $\square$ have their before-mentioned values.

In the particular case where $\mathrm{f}=0$, $\mathrm{g}=0$, $\mathrm{h}=0$, we have

$$A,\ B,\ C,\ F,\ G,\ H=\mathrm{bc},\ \mathrm{ca},\ \mathrm{ab},\ 0,\ 0,\ 0;\quad L^2=\mathrm{abc};$$

the equation $(\mathfrak{X})$ becomes

$$-2\mathrm{abc}(\mathrm{b}_{33}+\mathrm{c}_{22})+[\mathrm{c}(\mathrm{a}_1\mathrm{bc}+\mathrm{b}_1\mathrm{ca}+\mathrm{c}_1\mathrm{ab})+\mathrm{b}(\mathrm{a}_1\mathrm{bc}+\mathrm{b}_1\mathrm{ca}+\mathrm{c}_1\mathrm{ab})]-\mathrm{bc}(\mathrm{b}_1\mathrm{c}_2+\mathrm{c}_2\mathrm{a}_2+\mathrm{a}_3\mathrm{b}_3)=0,$$

that is,

$$-2\mathrm{abc}(\mathrm{b}_{33}+\mathrm{c}_{22})-\mathrm{bc}\,\mathrm{b}_2\mathrm{c}_1+\mathrm{ca}(\mathrm{b}_1\mathrm{c}_1+\mathrm{b}_3^2)+\mathrm{ab}(\mathrm{b}_2\mathrm{c}_3+\mathrm{c}_1^2)=0,$$

and the equation $(\mathfrak{f})$ becomes

$$4\mathrm{abc}\cdot\mathrm{a}_{23}-[\mathrm{a}_3(\mathrm{a}_2\mathrm{bc}+\mathrm{b}_2\mathrm{ca}+\mathrm{c}_2\mathrm{ab})+\mathrm{a}_2(\mathrm{a}_3\mathrm{bc}+\mathrm{b}_3\mathrm{ca}+\mathrm{c}_3\mathrm{ab})]$$
$$-\mathrm{ca}(\mathrm{a}_2\mathrm{b}_3-\mathrm{a}_3\mathrm{b}_2)-\mathrm{ab}(\mathrm{c}_2\mathrm{a}_3-\mathrm{c}_3\mathrm{a}_2)=0,$$

that is,

$$4\mathrm{abc}\cdot\mathrm{a}_{23}-2\mathrm{bc}\cdot\mathrm{a}_2\mathrm{a}_3-2\mathrm{ca}\cdot\mathrm{a}_2\mathrm{b}_3-2\mathrm{ab}\cdot\mathrm{a}_3\mathrm{b}_2=0.$$

Dividing by 2abc and 4abc respectively, and completing the system, the six equations are

$$\mathrm{b}_{33}+\mathrm{c}_{22}-\frac{\mathrm{b}_2^2}{2\mathrm{b}}-\frac{\mathrm{c}_3^2}{2\mathrm{c}}-\frac{\mathrm{bc}_1}{2\mathrm{a}}+\frac{\mathrm{bc}_2}{2\mathrm{b}}+\frac{\mathrm{bc}_3}{2\mathrm{c}}=0 \qquad (\mathfrak{A})$$
$$\mathrm{c}_{11}+\mathrm{a}_{33}-\frac{\mathrm{c}_3^2}{2\mathrm{c}}-\frac{\mathrm{a}_1^2}{2\mathrm{a}}-\frac{\mathrm{ca}_1}{2\mathrm{a}}+\frac{\mathrm{ca}_2}{2\mathrm{b}}+\frac{\mathrm{ca}_3}{2\mathrm{c}}=0 \qquad (\mathfrak{B})$$
$$\mathrm{a}_{22}+\mathrm{b}_{11}-\frac{\mathrm{a}_1^2}{2\mathrm{a}}-\frac{\mathrm{b}_2^2}{2\mathrm{b}}-\frac{\mathrm{ab}_1}{2\mathrm{a}}+\frac{\mathrm{ab}_2}{2\mathrm{b}}+\frac{\mathrm{ab}_3}{2\mathrm{c}}=0 \qquad (\mathfrak{C})$$
$$\mathrm{a}_{23}-\frac{\mathrm{a}_2\mathrm{a}_3}{2\mathrm{a}}-\frac{\mathrm{a}_2\mathrm{b}_3}{2\mathrm{b}}-\frac{\mathrm{a}_3\mathrm{c}_2}{2\mathrm{c}}=0 \qquad (\mathfrak{f})$$
$$\mathrm{b}_{31}-\frac{\mathrm{b}_3\mathrm{b}_1}{2\mathrm{b}}-\frac{\mathrm{b}_3\mathrm{a}_1}{2\mathrm{a}}-\frac{\mathrm{b}_1\mathrm{c}_3}{2\mathrm{c}}=0 \qquad (\mathfrak{g})$$
$$\mathrm{c}_{12}-\frac{\mathrm{c}_1\mathrm{c}_2}{2\mathrm{c}}-\frac{\mathrm{c}_1\mathrm{b}_2}{2\mathrm{b}}-\frac{\mathrm{c}_2\mathrm{a}_1}{2\mathrm{a}}=0. \qquad (\mathfrak{h})$$

These are in fact Lamé's equations, *Leçons sur les coordonnées curvilignes*, etc., Paris (1859), pp. 76, 78, viz. the first of the equations (8), p. 76, is

$$\frac{d^2H}{dp_2\,dp_3}=\frac{1}{H_2}\frac{dH_2}{dp_3}\frac{dH}{dp_2}+\frac{1}{H_3}\frac{dH_3}{dp_2}\frac{dH}{dp_3},$$

which, in the notation of the present paper, is

$$(\sqrt{\mathrm{a}})_{23}=\frac{1}{\sqrt{\mathrm{b}}}(\sqrt{\mathrm{b}})_3(\sqrt{\mathrm{a}})_2+\frac{1}{\sqrt{\mathrm{c}}}(\sqrt{\mathrm{c}})_2(\sqrt{\mathrm{a}})_3.$$

Here

$$(\sqrt{\mathrm{a}})_2=\tfrac{1}{2}\frac{\mathrm{a}_2}{\sqrt{\mathrm{a}}},$$
$$(\sqrt{\mathrm{a}})_{23}=\tfrac{1}{2}\left(\frac{\mathrm{a}_{23}}{\sqrt{\mathrm{a}}}-\tfrac{1}{2}\frac{\mathrm{a}_2\mathrm{a}_3}{\mathrm{a}\sqrt{\mathrm{a}}}\right);$$

and the equation therefore is

$$\frac{\mathrm{a}_{23}}{\sqrt{\mathrm{a}}}-\tfrac{1}{2}\frac{\mathrm{a}_2\mathrm{a}_3}{\mathrm{a}\sqrt{\mathrm{a}}}=\frac{1}{2}\frac{\mathrm{a}_2\mathrm{b}_3}{\sqrt{\mathrm{b}}\sqrt{\mathrm{a}}}+\tfrac{1}{2}\frac{\mathrm{a}_3\mathrm{c}_2}{\sqrt{\mathrm{c}}\sqrt{\mathrm{a}}},$$

which, multiplying by $\sqrt{\mathrm{a}}$, gives the foregoing equation

$$\mathrm{a}_{23}-\frac{\mathrm{a}_2\mathrm{a}_3}{2\mathrm{a}}-\frac{\mathrm{a}_2\mathrm{b}_3}{2\mathrm{b}}-\frac{\mathrm{a}_3\mathrm{c}_2}{2\mathrm{c}}=0.$$

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The first of the equations (9) p. 78 is

$$\frac{d}{dp_1}\left(\frac{1}{H_1}\frac{dH}{dp_1}\right)+\frac{d}{dp_2}\left(\frac{1}{H_2}\frac{dH_2}{dp_2}\right)=\frac{1}{H_2}\frac{dH_1}{dp_1}\frac{dH_1}{dp_2},$$

which, in the notation of the present paper, is

$$\left(\frac{(\sqrt{\mathrm{c}})_1}{\sqrt{\mathrm{a}}}\right)_1+\left(\frac{(\sqrt{\mathrm{c}})_2}{\sqrt{\mathrm{b}}}\right)_2=\frac{1}{\sqrt{\mathrm{a}}\sqrt{\mathrm{b}}}(\sqrt{\mathrm{a}})_2(\sqrt{\mathrm{b}})_1.$$

This gives first

$$\frac{(\sqrt{\mathrm{c}})_{11}}{\sqrt{\mathrm{a}}}-\frac{1}{2}\frac{(\sqrt{\mathrm{c}})_1\mathrm{a}_1}{\mathrm{a}\sqrt{\mathrm{a}}}+\frac{(\sqrt{\mathrm{c}})_{22}}{\sqrt{\mathrm{b}}}-\frac{1}{2}\frac{(\sqrt{\mathrm{c}})_2\mathrm{b}_2}{\mathrm{b}\sqrt{\mathrm{b}}}=\frac{1}{\sqrt{\mathrm{a}}\sqrt{\mathrm{b}}}(\sqrt{\mathrm{a}})_2(\sqrt{\mathrm{b}})_1,$$

and then

$$\frac{1}{2}\frac{\mathrm{c}_{11}}{\sqrt{\mathrm{c}}\sqrt{\mathrm{a}}}-\frac{1}{4}\frac{\mathrm{c}_1^2}{\mathrm{c}\sqrt{\mathrm{c}}\sqrt{\mathrm{a}}}-\frac{1}{4}\frac{\mathrm{c}_1\mathrm{a}_1}{\mathrm{a}\sqrt{\mathrm{c}}\sqrt{\mathrm{a}}}+\frac{1}{2}\frac{\mathrm{c}_{22}}{\sqrt{\mathrm{c}}\sqrt{\mathrm{b}}}-\frac{1}{4}\frac{\mathrm{c}_2^2}{\mathrm{c}\sqrt{\mathrm{c}}\sqrt{\mathrm{b}}}-\frac{1}{4}\frac{\mathrm{c}_2\mathrm{b}_2}{\mathrm{b}\sqrt{\mathrm{c}}\sqrt{\mathrm{b}}}=\frac{1}{4}\frac{\mathrm{a}_2\mathrm{b}_1}{\sqrt{\mathrm{a}}\sqrt{\mathrm{b}}},$$

which, on multiplying by $\sqrt{\mathrm{a}}\sqrt{\mathrm{b}}$ and reducing, gives the foregoing equation

$$\mathrm{a}_{22}+\mathrm{b}_{11}-\frac{\mathrm{a}_1^2}{2\mathrm{a}}-\frac{\mathrm{b}_2^2}{2\mathrm{b}}-\frac{\mathrm{ab}_1}{2\mathrm{a}}+\frac{\mathrm{ab}_2}{2\mathrm{b}}-\frac{\mathrm{a}_2\mathrm{b}_1+\mathrm{a}_1\mathrm{b}_2}{2\mathrm{c}}=0.$$

800.

NOTE ON THE STANDARD SOLUTIONS OF A SYSTEM OF LINEAR EQUATIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 38–40.]

To fix the ideas, the equations are assumed to be without constant terms. Supposing the system to be insufficient for the determination of the ratios of the unknown quantities, then regarding the unknown quantities as having a definite order of arrangement, there are certain solutions which may be regarded as standard solutions. Take the unknown quantities to be A, B, C, D, E, F, G, &c.; then assuming $A=0$, or else A, B, each $=0$, or else A, B, C, each $=0$, as many equations as possible, the system as thus modified will have a definite solution; for instance, the assumed equations A, B, C, D, E, each $=0$, may give a definite solution in which F is not $=0$, and it may then for convenience be put $=1$. We have thus a solution beginning with $F=1$. This being so, there will be a solution or solutions with $F=0$; we cannot then have A, B, C, D, E, each $=0$, but we again assume $A=0$, or else A, B, each $=0$, as many equations as possible; suppose A, B, C, each $=0$ give a definite solution, with D not $=0$; and then taking it for convenience to be $=1$, we have a solution beginning $D=1$, and for which $F=0$. Going on in this manner we obtain, it may be, a solution beginning $B=1$, and for which $D=0$, $F=0$; and so on, the process stopping, if not sooner, with a solution beginning $A=1$, and with the initial letters of the preceding solutions, each $=0$. We have in this manner the system of standard solutions, of a form such as

A	B	C	D	E	F	G	
0	0	0	0	0	1	$*$	
0	0	0	1	$*$	0	$*$	3—2
0	1	$*$	0	$*$	0	$*$	
1	0	$*$	0	$*$	0	$*$	

where the $*$ denotes a value which is not in general $= 0$, but which may in any particular case happen to be so.

For instance, let it be required for the binary quartic $(a, b, c, d, e \rangle x, y)^4$, to find the asyzygetic seminvariants of the degree 4 and weight 6. Assuming for the seminvariant the value in the left-hand column of the diagram, the unknown coefficients being A, B, C, D, E, F, G, this must be reduced to zero by the operation

$$\Delta = a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e;$$

and we thus obtain as many equations as there are terms of the degree 4 and weight 5, as appearing by the second column

		A	B	C	D	E	F	G
a^2e^2	$ad^2b + 2\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
A	$= a^2ce$	1						
B	$= ab^2e$		1					
C	$= abcd$			1				
D	$= ac^3$				1			
E	$= b^3d$					1		
F	$= b^2c^2$						1	

[The table on book p. 20 records, for each of the seven monomials in the diagram, the corresponding column giving the result of Δ acting upon it; the resulting standard solutions $A = 1, B = -1, \ldots, F = 1$ enumerate the irreducible composites P and Q used on p. 21.]

As is known, there is no irreducible solution, but only the composite forms

$$\begin{aligned}\mathrm{I} &= a(ace - ad^2 - b^2e + 2bcd - c^3),\\ \mathrm{II} &= (ac - b^2)(ae - 4bd + 3c^2),\end{aligned}$$

the developed values of which are given above: II (as a form beginning $A = 1$ and with $B = 0$) can be nothing else than, and is in fact $= Q$: and so I (as a form beginning with $A = 1$, $B = -1$) can be nothing else than, and is in fact $= Q - P$; that is, we have

$$\begin{aligned}P &= -\mathrm{I} + \mathrm{II}, &\qquad \text{or}\quad \mathrm{I} &= -P + Q,\\ Q &= \qquad\ \mathrm{II}, & \mathrm{II} &= \qquad\ \ Q;\end{aligned}$$

and so, in general, we have a standard set of values for the asyzygetic seminvariants of a given degree and weight; or, what is the same thing, for the covariants of a given deg-order.

801.

ON SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 131–138.]

The present paper is a somewhat fragmentary one, but it contains some results which seem to me to be worth putting on record.

I consider here not any binary quantic in particular, but the whole series $(a, b, c\,\rangle x, y)^2$, $(a, b, c, d\,\rangle x, y)^3$, &c.; or in a somewhat different point of view, I consider the indefinite series of coefficients $(a, b, c, d, e, \ldots)$; here, instead of covariants and invariants, we have only seminvariants; viz. a seminvariant is a function reduced to zero by the operator

$$\Delta = a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e + 5e\partial_f + \ldots,$$

for instance, seminvariants are

$$a, \quad ac - b^2, \quad a^2d - 3abc + 2b^3, \quad a^2d^2 + 4ac^3 + 4b^3d - 6abcd - 3b^2c^2,$$

$$ae - 4bd + 3c^2, \quad ace - ad^2 - b^2e + 2bcd - c^3, \quad \&c.$$

A seminvariant is of a certain degree θ in the coefficients, and of a certain weight w (viz. the coefficients a, b, c, d, ... are reckoned as being of the weights 0, 1, 2, 3, ... respectively); it is, moreover, of a certain rank ρ; viz. according as the highest letter therein is a, c, d, e, ... (it is never b), the rank is taken to be 0, 2, 3, 4, ..., and we have $w = \frac{1}{2}\rho\theta$. The seminvariant... may be regarded as belonging to a quantic $(a, \ldots\rangle x, y)^n$, the order of which is n, equal to or greater than ρ; viz. in regard to such quantic the seminvariant, say A, is the leading coefficient of a covariant

$$(A, B, \ldots, K\,\rangle x, y)^{n\theta - 2w},$$

where the weights of the successive coefficients are w, $w+1$, ... up to $n\theta - w$; hence the number of terms less unity, that is, μ, is $= n\theta - 2w$; the least value of μ is thus $= \rho\theta - 2w$, which is either zero, or positive; in the former case, $w = \frac{1}{2}\rho\theta$, the seminvariant is an invariant of the quantic $(a, \ldots\rangle x, y)^\rho$, the order of which is equal to the rank of the seminvariant; but if $w < \frac{1}{2}\rho\theta$, then it is the leading coefficient of a covariant $(A, B, \ldots, E\,\rangle x, y)^{\rho\theta - 2w}$ of the same quantic $(a, \ldots\rangle x, y)^\rho$; and in every case, taking $n > \rho$, the seminvariant is the leading coefficient A of a covariant $(A, B, \ldots, K\,\rangle x, y)^{n\theta - 2w}$ of a quantic $(a, \ldots\rangle x, y)^n$.

Take A as belonging to the quantic $(a, \dots \rangle x, y)^n$; corresponding to such quantic, we have an operator Λ of the same rank n, viz.

$$\begin{aligned} \Lambda &= 2b\partial_a + c\partial_b && \text{for } n = 2, \\ &= 3b\partial_a + 2c\partial_b + d\partial_c && n = 3, \\ &= 4b\partial_a + 3c\partial_b + 2d\partial_c + e\partial_d && n = 4, \end{aligned}$$

Operating with Λ on A, we have a series of terms

$$A,\ \Lambda A,\ \Lambda^2 A,\ \dots,\ \Lambda^{n\theta-2w} A,$$

but the next term $\Lambda^{n\theta-2w+1}A$, and of course every succeeding term, is $= 0$, and this being so, the coefficients of the covariant $(A, B, \dots, K \rangle x, y)^{n\theta-2w}$ are

$$\left(1,\ \frac{1}{1}\Lambda,\ \frac{1}{1 \cdot 2}\Lambda^2,\ \dots\right) A,$$

or what is the same thing, each coefficient is obtained from the next preceding one by the formulae

$$B = \frac{1}{1}\Lambda A, \qquad C = \frac{1}{2}\Lambda B, \qquad D = \frac{1}{3}\Lambda C, \dots.$$

The coefficients A and K, B and J, ... are derived one from the other by reversal of the order of the coefficients of $(a, b, \dots \rangle x, y)^n$, with or without a change of sign, and thus it is only necessary to calculate up to the middle coefficient, or pair of coefficients; and we obtain, moreover, a verification.

Calculating in this manner the covariant

$$(A, B, \dots, K \rangle x, y)^{n\theta-2w}$$

which belongs to the quantic $(a, \dots \rangle x, y)^n$, if we herein change a, b, c, ... into $ax + by$, $bx + cy$, $cx + dy$, ... we obtain the covariant belonging to the quantic $(a, \dots \rangle x, y)^{n+1}$; and in this covariant making the like change, or what is the same thing, in the first-mentioned covariant changing a, b, c, ... into $(a, b, c \rangle x, y)^2$, $(b, c, d \rangle x, y)^2$, $(c, d, e \rangle x, y)^2$, ... we have the covariant belonging to $(a, \dots \rangle x, y)^{n+2}$; and in like manner we obtain the covariant belonging to the quantic $(a, \dots \rangle x, y)^n$ of any given order n.

In particular, if $w = \frac{1}{2}\rho\theta$, that is, if the given seminvariant be an invariant of $(a, \dots \rangle x, y)^\rho$, then we obtain the series of covariants directly from A, by therein changing a, b, c, ... into $ax + by$, $bx + cy$, $cx + dy$, ... and in the result making the like change; or what is the same thing, in A changing a, b, c, ... into $(a, b, c \rangle x, y)^2$, $(b, c, d \rangle x, y)^2$, $(c, d, e \rangle x, y)^2$, ... and so on until we obtain the covariant for the quantic $(a, \dots \rangle x, y)^n$ of the given order n.

A seminvariant which cannot be expressed as a rational and integral function of lower seminvariants is said to be irreducible. The theory is distinct from that of the irreducible covariants of a quantic of a given order; for instance, as regards the cubic $(a, b, c, d \rangle x, y)^3$, we have the irreducible covariant (invariant)

$$a^2d^2 + 4ac^3 + 4b^3d - 6abcd - 3b^2c^2,$$

but this is not an irreducible seminvariant; it is

$$= (ac - b^2)(ae - 4bd + 3c^2) - a(ace - ad^2 - b^2e + 2bcd),$$

or, what is the same thing, there is not for the quartic $(a, b, c, d, e \rangle x, y)^4$ or for the higher quantics, any irreducible covariant having this for the leading coefficient.

We may consider the question to determine the number of asyzygetic seminvariants of a given degree and weight. For instance, taking the weights up to 12, so that the series of letters extends as far as m, then for the degrees 1, 2, 3 we have as follows:

$w =$	0	1	2	3	4	5	6	7	8	9	10	11	12
Deg. 1	a	b	c	d	e	f	g	h	i	j	k	l	m
Num.	1	1	1	1	1	1	1	1	1	1	1	1	1
Diff.	0	0	0	0	0	0	0	0	0	0	0	0	0
Deg. 2	a^2	ab	ac	ad	ae	af	ag	ah	ai	aj	ak	al	am
		b^2	bc	bd	be	bf	bg	bh	bi	bj	bk	bl	
			c^2	cd	ce	cf	cg	ch	ci	cj	ck	cl	
				d^2	de	df	dg	dh	di	dj	dk		
					e^2	ef	eg	eh	ei	ej	ek		
						f^2	fg	fh	fi	fj			
							g^2	gh	gi	gj			
								h^2	hi	ij			
									i^2	ij			
Num.	1	1	2	2	3	3	4	4	5	5	6	6	7
Diff.	1	0	1	0	1	0	1	0	1	0	1	0	1
Deg. 3	a^3	a^2b	a^2c	a^2d	a^2e	a^2f	a^2g	a^2h	a^2i	a^2j	a^2k	a^2l	a^2m
		ab^2	abc	abd	abe	abf	abg	abh	abi	abj	abk	abl	
			b^3	ac^2	acd	ace	acf	acg	ach	aci	acj	ack	
					b^2d	b^2e	ad^2	ade	adf	adg	adh	adi	
					bc^2	bcd	ae^2	b^2h	aef	aeg	aeh		
						c^3	bce	bch	b^2i	b^2j			
							bd^2	bdf	bcg	bdi			
							c^2d	bef	cd^2	cei			
								cde	ce^2	def			
								d^2e		d^2g			
Num.	1	1	2	3	4	5	7	8	10	12	14	16	19
Diff.	1	0	1	1	1	1	2	1	2	2	2	2	3

For the degree 1, the line of differences shows that the only seminvariant is ($W = 0$), the seminvariant a.

For the degree 2, the line of differences 0, 1, 1, 0, . . ., shows that the number of seminvariants is $= 1$ for each even degree, $= 0$ for each odd degree; thus for the weight there is a seminvariant $= a^2$ which of course is not, irreducible; while for each of the other even weights we have a single irreducible seminvariant; as is well known, the forms are

$W =$	2	4	6	8	10	
	$ac - b^2$	$ae - 4bd + 3c^2$	$ag - 6bf + 15ce - 10d^2$	$ai - 8bh + 28cg - 56df + 35e^2$	$ak - \dots$	$am - 12bl + 6$

For degree 3, the line of differences shows that the

W	$= 0$	1	2	3	4	5	6	7	8	9	10	11	12
Nos. irr. $=$	0	1	1	1	1	1	2	1	2	2	2	2	3

but inasmuch as for each even weight there is a quadric seminvariant, which multiplied by a given a cubic seminvariant, to obtain the number of irreducible cubic seminvariants we subtract

1	0	1	0	1	0	1	0	1	0	1	0	1
0	0	0	1	0	1	1	1	1	2	1	2	2

or the numbers of irreducible cubic seminvariants are as in the line last written down.

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There is a convenience however in giving, for each even weight, as well the rejected reducible covariant; and the entire series of results is found to be

$W =$	0	2	4	6	8	10	12	14	16	18	20	22	24
a^2+1	$ac+1$	a^2d+1	ad^2+1	ae^2+1	af^2+1	ag^2+1	ah^2+1	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	
$\vdots$													

[Book p. 26 (PDF p. 46) gives an extensive multi-column tabulation, indexed by even weights from $W = 0$ up to $W = 12$ and beyond, of the asyzygetic and irreducible cubic seminvariants with the integer multipliers needed for their reductions. The verifications carried out by Cayley involve degrees 1, 2, 3 cubic seminvariants whose entries are sums of monomials in $a, b, c, \dots$ with small integer coefficients, in the format shown above.]

The canonical form given for the quintic in my Tenth Memoir on Quantics [693] belongs to a series, viz. writing now the small roman letters (instead of the italic letters) for the series of coefficients, and using the italic letters a, c, d, e, f, ... to denote seminvariants, they are as follows:

$$\begin{array}{ll}
0 & a = \mathrm{a}, \\
2 & c = \mathrm{ac} - \mathrm{b}^2, \\
3 & d = \mathrm{a}^2\mathrm{d} - 3\mathrm{abc} + 2\mathrm{b}^3 \quad (= f \text{ in the tenth memoir}), \\
4 & e = \mathrm{ae} - 4\mathrm{bd} + 3\mathrm{c}^2 \quad (= b \text{ in the tenth memoir}), \\
5 & f = \mathrm{a}^2\mathrm{f} - 5\mathrm{abe} + 2\mathrm{acd} + 8\mathrm{b}^2\mathrm{d} - 6\mathrm{bc}^2, \\
6 & g = \mathrm{ag} - 6\mathrm{bf} + 15\mathrm{ce} - 10\mathrm{d}^2, \\
7 & h = \mathrm{a}^2\mathrm{h} - 7\mathrm{abg} + 9\mathrm{acf} - 5\mathrm{ade} + 12\mathrm{b}^2\mathrm{f} - 30\mathrm{bce} + 20\mathrm{bd}^2, \\
8 & i = \mathrm{ai} - 8\mathrm{bh} + 28\mathrm{cg} - 56\mathrm{df} + 35\mathrm{e}^2, \\
 & \&\mathrm{c}.
\end{array}$$

Writing also (instead of d in the tenth memoir)

$$\bar{e} = \mathrm{ace} - \mathrm{ad}^2 - \mathrm{b}^2\mathrm{e} + 2\mathrm{bcd} - \mathrm{c}^3,$$

so that the equation $a^2d^2 + a^2bc - 4c^3 \cdot f = 0$ of the tenth memoir is in the present notation $a^2\bar{e} - a^2ec + 4c^3 - d^2 = 0$, then the series of canonical forms is

$$\begin{array}{rl}
\text{Quadric} & (1, 0, c \,\rangle x, y)^2, \\
\text{Cubic} & (1, 0, c, d \,\rangle x, y)^3, \\
\text{Quartic} & (1, 0, c, d, a^2e - 3c^2 \,\rangle x, y)^4, \\
\text{Quintic} & (1, 0, c, d, a^2e - 3c^2, a^2f - 2cd \,\rangle x, y)^5, \\
 & \&\mathrm{c}.
\end{array}$$

the series of coefficients being

$$\begin{array}{lllllllll}
1, & 0, & c, & d, & a^2e + 1, & a^2f + 1, & a^2g + 1, & a^2h + 1, & a^2i + 1, \\
 & & & & c^2 - 3, & cd - 2, & a^2ce - 15, & & \\
 & & & & c^3 + 45, & a^2de + 5, & a^4e^2 - 35, & & \\
 & & & & d^2 + 10, & c^2d + 3, & a^2c^2e + 630, & & \\
 & & & & & & d^2f + 56, & & \\
 & & & & & & c^4 - 1575, & & \\
 & & & & & & cd^2 - 392 & &
\end{array}$$

these values being, in fact, the expressions in terms of the seminvariants a, c, d, &c. of

$$1, \quad 0, \quad \mathrm{ac}, \quad \mathrm{a}^2\mathrm{d}, \quad \mathrm{a}^3\mathrm{e}, \quad \mathrm{a}^4\mathrm{f}, \quad \mathrm{a}^5\mathrm{g}, \quad \mathrm{a}^6\mathrm{h}, \quad \mathrm{a}^7\mathrm{i}.$$

I annex verifications of the foregoing values. The original page tabulates, for the canonical-form coefficients a^2e-3c^2, a^2f-2cd, $a^2g-15a^2ce+45c^3+10d^2$, $a^2h-28a^2cg+35a^2df-630ab^2f+112c^2e+\ldots$, the integer multipliers attached to each monomial a^2e, abd, ac^2, b^2c, a^2f, abe, $abcd$, ac^3, a^2g, abf, acf, ade, b^3, b^2d, b^2e, bce, bcf, bd^2, c^2d, c^2e, c^3, cd^2, d^2, …, displayed as a sequence of small tables side by side. The first such block (for a^2e-3c^2) is

	a^2e	$-3c^2$
a^2e	$+1$	
abd	-4	
ac^2	$+3$	-3
b^2c		$+8$

	a^2f	$-2cd$
a^2f	$+1$	
a^2cd		-3
abe	-5	
$abcd$	-2	
ac^3		$+10$

	a^2g	$-15a^2ce$	$+45c^3$	$+10d^2$
a^2g	$+1$			
abf	-6			
acf		-15		
ade		-10		
abe	$+5$			
b^3			$+45$	
b^2d			-100	$+10$
b^2e				
bce			$+45$	
bd^2			-30	$+5$
c^2e			-15	
c^3			$+45$	
cd^2			-30	$+5$
d^2				$+6$

The remaining block, for $a^2h-28a^2cg+35a^2df-630ab^2f+112c^2e+\ldots$, follows the same layout; the entries continue on book p. 29.

	a^2i	$-36a^2bh$	$-56a^2dg$	$+35a^2e^2$	$+630a^2cf$	$-1575c^4$
ai	$+1$					
abh	-8					
ach						
acg	$+28$	-36				
adg	-56		$+56$			
adf	-56					
ae^2	$+35$			$+35$		
b^2g		$+290$				
b^2h						
bcg		-460	$+1200$			
bdf		$+840$	-840			
be^2			-315	$+112$		
c^2g		$+1890$			$+630$	-1575
cd^2			$+1080$			
ce^2				$+672$	-315	
d^2f					$+6300$	-6275
de^2						$+1004$

It would be interesting to obtain the general law for the expressions of the canonical coefficients in terms of the seminvariants.

802.

NOTE ON CAPTAIN MACMAHON'S PAPER, "ON THE DIFFERENTIAL EQUATION $X^{-\frac{1}{3}}dx + Y^{-\frac{1}{3}}dy + Z^{-\frac{1}{3}}dz = 0$."

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 182–184.]

In general, if $f = (x, y, 1)^3, = 0$ be the equation of a cubic curve, and if

$$d\omega = \frac{dx}{\frac{df}{dy}} = -\frac{dy}{\frac{df}{dx}},$$

then if 1, 2, 3 are the intersections of the curve by an arbitrary right line, the coordinates of these points being (x_1, y_1), (x_2, y_2), (x_3, y_3) respectively, we have, by Abel's theorem,

$$d\omega_1 + d\omega_2 + d\omega_3 = 0,$$

viz. this is the differential relation corresponding to the integral relation which expresses that the three points are the intersections of the cubic curve by a right line, or say to the integral equation

$$\nabla, = \begin{vmatrix} y_1, & y_2, & y_3 \\ x_1, & x_2, & x_3 \\ 1, & 1, & 1 \end{vmatrix} = 0,$$

in which equation y_1, y_2, y_3 are regarded as functions of x_1, x_2, x_3 respectively, given by means of the equations $f_1 = 0$, $f_2 = 0$, $f_3 = 0$, which express that the points are on the cubic curve. See my "Memoir on the Abelian and Theta Functions," [819].

In particular, if the equation of the curve is

$$f, = \tfrac{1}{3}y^3 - (A + 3Bx + 3Cx^2 + Dx^3), = \tfrac{1}{3}(y^3 - X), = 0,$$